
Semi-Structured Correlation-aware Weight Sparsity (CWS) for Efficient LLM Inference

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Abstract

1 Extreme weight pruning can yield substantial reductions in model size, yet standard
2 pruning methods scoring each weight in isolation miss an important class of redun-
3 dancy. Groups of correlated input channels whose mixed-sign weights destructively
4 cancel in the layer output. We propose **correlation-aware weight sparsity (CWS)**,
5 a post-training pruning method that identifies and removes these groups. These
6 *cancellation groups* are invisible to standard per-weight saliency scores. CWS
7 detects them by deriving its selection criterion from the full activation covariance
8 through greedy Schur complement updates within fixed-width column blocks, then
9 applies exact Optimal Brain Surgeon weight corrections after each selection step.
10 Across HGRN-1.3B, a 1B FLA transformer, and LLaMA-7B, CWS consistently
11 improves perplexity over existing one-shot pruning methods at moderate and high
12 sparsity levels. The method is particularly effective in the sub-2B scale, where
13 exact second-order correction remains critical for maintaining model quality. We
14 further show that the resulting sparsity patterns on vectorized CPU implementations
15 with AVX-512 support approach $2\times$ end-to-end speedup over dense baselines, with
16 the gain fully preserved after post-pruning recovery.

1 Introduction

Post-training pruning methods for large language models (LLMs) typically score weights independently. Prior post-training pruning methods, have made substantial progress but share a blind spot in their selection criterion. Magnitude [Han et al., 2015], Wanda [Sun et al., 2023], RIA [Zhang et al., 2024], and SparseGPT [Frantar and Alistarh, 2023] all score weights individually, assessing the marginal cost of removing each weight in isolation. AWP [Liu et al., 2025] searches more broadly through iterative gradient projection, but still has no explicit mechanism to identify correlated groups. When two input channels are positively correlated and their weights carry opposite signs, their contributions to the layer output partially cancel. Their joint pruning cost is lower than either individual score predicts, and the pair is a natural candidate for removal. These *cancellation groups* are invisible to all existing selection criteria.

We propose **correlation-aware weight sparsity (CWS)**, which makes them visible by deriving its selection criterion from the full activation covariance rather than its diagonal, then applies Optimal Brain Surgeon (OBS) weight corrections [Hassibi and Stork, 1992]. The method operates post-training, requires no gradient computation, and produces semi-structured sparsity patterns, compatible with efficient LLM inference.

We evaluate across three model families spanning two architectures and two scales, and provide hardware benchmarks on CPU with AVX-512 sparse kernels. Our main contributions are:

- **Cancellation-aware pruning.** CWS outperforms SparseGPT at every sparsity level from 30% to 80% on sub-2B models, with gains ranging from $1.007\times$ at 30% to $2.06\times$ at 80% on the 1B transformer. The method generalizes across both attention-based and gated recurrent SSM architectures, confirming that the gains arise from activation covariance structure rather than architectural specifics.
- **Scale and architecture evaluation.** We evaluate post-training pruning at billion-parameter scale on both transformer and recurrent SSM architectures, including LLaMA-7B and HGRN-1.3B. The HGRN-1.3B is included to demonstrate the effect of weight sparsity on end-to-end inference, where a larger fraction of inference cost is placed in feed forward layers rather than attention.
- **CPU inference benchmarks.** We demonstrate that the sparsity patterns produced by CWS translate directly to inference throughput on commodity hardware. Using custom AVX-512 kernels tuned to the semi-structured sparsity format, pruned HGRN-1.3B models achieve up to $1.95\times$ end-to-end speedup over dense BF16 baselines, with no specialized accelerator required.

2 Methods

2.1 Problem Formulation

We prune each linear layer independently. Consider a weight matrix $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, where each row corresponds to one output neuron and each column corresponds to one input channel. The goal is to remove a fixed fraction of weights while preserving the layer output on representative calibration data.

We use semi-structured sparsity. Instead of pruning an arbitrary number of weights across the full matrix, we enforce the same sparsity ratio within every row. In practice, this means that each output neuron keeps the same number of non-zero input connections. This produces a regular sparsity pattern that is easier to accelerate with sparse kernels than fully unstructured pruning.

Post training pruning applies a binary mask $\mathbf{M} \in \{0, 1\}^{d_{\text{out}} \times d_{\text{in}}}$ to the weight matrix:

$$\hat{\mathbf{W}} = \mathbf{M} \odot \mathbf{W},$$

where entries of $\mathbf{M}$ are initialized to one and set to zero when the corresponding weights are pruned. For target sparsity ratio s , each row removes exactly $k = \lfloor d_{\text{in}} \cdot s \rfloor$ weights.

For output neuron i , the pruning problem reduces to selecting which subset of weights to remove. Let $S \subset [d_{\text{in}}]$ denote the indices of the pruned weights. The objective is to minimize the change in the layer output under the activation distribution observed during calibration:

$$\min_{\substack{S \subseteq [d_{\text{in}}] \\ |S|=k}} \mathbb{E}_x \left[\left(\mathbf{w}_i^\top x - (\mathbf{M}_i \odot \mathbf{w}_i)^\top x \right)^2 \right] = \mathbf{w}_{i,S}^\top \boldsymbol{\Sigma}_X[S, S] \mathbf{w}_{i,S}, \quad (1)$$

66 where $\boldsymbol{\Sigma}_X = \mathbb{E}[xx^\top]$ is the input activation covariance matrix.

67 Equation 1 shows that pruning error depends not only on the magnitude of the removed weights,
68 but also on correlations between their input activations. If two correlated channels contribute with
69 opposite signs, their outputs partially cancel, reducing the cost of pruning them together. Standard
70 per weight scoring methods ignore this interaction because they only use diagonal statistics.

71 2.2 Activation Statistics

72 All methods require statistics of the input activations. We register forward hooks on every linear
73 layer and collect a calibration set of 64 batches (batch size 4, sequence length 512) drawn from C4.
74 For methods that only need the diagonal of $\boldsymbol{\Sigma}_X$ (Wanda, RIA), we accumulate the per channel RMS
75 activation:

$$\hat{\sigma}_j = \sqrt{\frac{1}{N} \sum_{n=1}^N x_{n,j}^2}. \quad (2)$$

76 For methods requiring the full covariance (CWS, SparseGPT), we accumulate $\mathbf{H} = \frac{2}{N} \mathbf{X}^\top \mathbf{X}$ (the
77 empirical Hessian of the squared output error), then compute $\mathbf{H}^{-1}$ via Cholesky decomposition with
78 a damping term $\lambda = 0.01 \cdot \text{mean}(\text{diag}(\mathbf{H}))$ for numerical stability.

79 2.3 Cancellation-Aware Pruning: CWS

80 The objective in Equation (1) is a quadratic form in the pruned weights. When two input channels j, j'
81 are positively correlated ($\boldsymbol{\Sigma}_X[j, j'] > 0$) and their weights carry opposite signs ($W_{i,j} \cdot W_{i,j'} < 0$),
82 their joint contribution to the layer output is smaller than either per-weight score predicts. Standard
83 diagonal scoring (Wanda, RIA, SparseGPT) is blind to these cancellation groups.

84 **Greedy OBS residual selection.** We solve the combinatorial selection problem greedily. After
85 pruned set S' has been chosen, the marginal cost of additionally pruning column j is:

$$\delta(j \mid S') = \frac{r_j^2}{d_j}, \quad (3)$$

86 where r_j and d_j are the j -th residual and diagonal element of the evolving Cholesky factor of $\mathbf{H}^{-1}$,
87 updated via rank-1 Schur complement steps after each selection.

88 When no weights have yet been pruned ($S' = \emptyset$), the criterion reduces exactly to the SparseGPT
89 saliency score. As pruning progresses, the residual updates incorporate correlations between previ-
90 ously pruned and candidate weights, enabling the method to identify the cancellation groups that
91 diagonal methods cannot capture.

92 **Block-level variant.** On models with $d_{\text{in}} \geq 2048$, global greedy selection breaks down for two
93 reasons: (1) the selected mask is not column-ordered, violating the assumption of the subsequent OBS
94 correction; and (2) thousands of Schur complement rank-1 updates cause the residual diagonal to drift
95 numerically even in float64. We resolve both by restricting selection and correction to fixed width
96 blocks of $B = 128$ columns. Within each block, $\lfloor B \cdot s \rfloor$ CWS steps select weights to prune from
97 the block’s $B \times B$ submatrix of $\mathbf{H}^{-1}$; OBS corrections are applied immediately after. Blocks are
98 processed left to right, eliminating the ordering mismatch and capping numerical drift at B updates
99 per block.

100 On the 1B model, where global CWS is numerically stable, the block variant captures slightly less
101 cross-block cancellation (PPL 25.5 vs. 24.1 at 50% sparsity); on LLaMA-7B and HGRN-1.3B, it is
102 strictly superior to the global variant.

3 Experiments

3.1 Experimental Setup

Models. We evaluate on three model families spanning two architectures and two scales: (1) a 1B-parameter FLA transformer, (2) LLaMA-7B [Touvron et al., 2023] (openlm-research/open_llama_7b), and (3) HGRN-1.3B [Qin et al., 2023] (fla-hub/hgrn-1.3B-100B), a gated recurrent SSM with no attention layers. Including both attention-based and recurrent architectures tests whether gains from CWS are architecture-agnostic. All methods are calibrated on 64 batches (batch size 4, sequence length 512) drawn from C4 [Raffel et al., 2020].

Baselines. We compare CWS against Wanda [Sun et al., 2023], RIA [Zhang et al., 2024], SparseGPT [Frantar and Alistarh, 2023], and AWP [Liu et al., 2025]. We also include two intermediate ablations: greedy covariance pruning without weight correction (GREEDY-COV), and greedy covariance pruning with closed-form batch correction (GREEDY+CORR). These isolate the contribution of the selection criterion step.

Evaluation. Perplexity is measured on the WikiText-2 test set [Merity et al., 2016] using non-overlapping 2048-token chunks Frantar and Alistarh [2023]. Downstream accuracy uses zero-shot lm-evaluation-harness [Gao et al., 2024] on ARC-Easy, ARC-Challenge, HellaSwag, PIQA, WinoGrande, and LAMBADA. We report the unweighted average (**Avg**). All models pruned one-shot and loaded in bfloat16.

3.2 TinyLlama-1.1B

Table 1: TinyLlama-1.1B at 50% sparsity: task accuracy breakdown. PPL on WikiText-2 test set. Downstream accuracies from zero-shot lm-eval. All methods calibrated on WikiText-2.

Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
Dense baseline	7.8	.557	.314	.592	.730	.589	.556
Wanda	10.7	.495	.271	.497	.676	.551	.498
RIA	10.8	.497	.270	.500	.678	.560	.501
SparseGPT	11.8	.483	.288	.518	.684	.575	.510
AWP	10.7	.496	.271	.497	.679	.551	.499
CWS	10.2	.487	.272	.508	.682	.554	.501

Table 2: TinyLlama-1.1B at 80% sparsity: task accuracy breakdown. PPL on WikiText-2 test set. Downstream accuracies from zero-shot lm-eval.

Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
Dense baseline	7.8	.557	.314	.592	.730	.589	.556
Wanda	615	.274	.242	.272	.523	.502	.363
RIA	902	.268	.258	.268	.515	.502	.362
SparseGPT	1490	.287	.237	.274	.516	.510	.365
AWP	637	.271	.242	.271	.517	.509	.362
CWS	217.27	.282	.236	.285	.522	.502	.365

3.3 HGRN-1.3B

HGRN-1.3B [Qin et al., 2023] is a gated recurrent SSM with no attention layers. We include it alongside transformer architectures because recurrent SSMs place a larger fraction of inference cost in feed forward layers, where weight sparsity directly reduces computation. This makes HGRN-1.3B a useful setting for studying the effect of semi-structured pruning independent of attention sparsity. Tables 4 and 12 report results at 50% and 80% sparsity. Table 6 and As shown in Figure 2 illustrate the full sparsity sweep from 30% to 80%. All accuracies are zero shot.

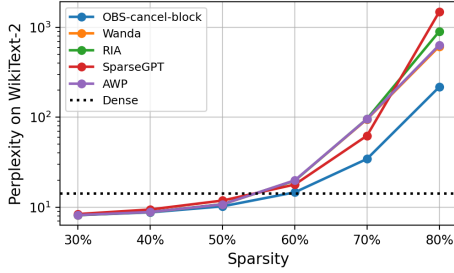


Figure 1: Sparsity-vs-perplexity. Sparsity sweep on TinyLlama-1.1B evaluated on WikiText-2.

Sparsity	CWS	Wanda	RIA	SGPT	AWP
30%	8.13	8.15	8.14	8.37	8.16
40%	8.74	8.82	8.81	9.41	8.82
50%	10.20	10.73	10.77	11.83	10.73
60%	14.60	19.76	19.69	17.83	19.86
70%	34.40	94.92	96.21	62.38	95.80
80%	217.27	615	902	1,490	637

Table 3: Sparsity sweep on the TinyLlama-1.1B. PPL on WikiText-2 test set. The lower PPL values are favorable.

Table 4: HGRN-1.3B at 50% sparsity. PPL on WikiText-2. Downstream accuracies from zero-shot lm-eval.

Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	LMBD	Avg↑
Dense baseline	11.8	.510	.275	.480	.712	.528	.383	.481
Wanda	350	.312	.263	.295	.545	.517	.003	.323
RIA	348	.305	.255	.298	.542	.511	.004	.319
SparseGPT	17.4	.467	.264	.433	.676	.519	.215	.429
AWP	426.2	.327	.241	.293	.525	.530	.014	.322
CWS	13.88	.482	.264	.443	.678	.501	.286	.474

CWS achieves the lowest perplexity at every sparsity level on HGRN-1.3B, with its margin over SparseGPT growing from $1.007\times$ at 30% to $1.067\times$ at 50% and $1.430\times$ at 80%.

Uncorrected methods collapse at 50% sparsity across both sub-2B architectures. At 50% sparsity, only CWS and SparseGPT retain meaningful perplexity and task accuracy across both sub-2B architectures, while uncorrected or approximately corrected methods collapse. This highlights the importance of exact second-order correction in the sub-2B regime.

3.4 Post-pruning Recovery at 80% Sparsity

At 80% sparsity, zero-shot CWS reaches PPL 4,865 with average task accuracy 0.301. A brief post-pruning training pass recovers meaningful accuracy without modifying the sparsity mask. Table 7 compares three recovery strategies applied to the CWS 80% pruned model: standard fine-tuning, knowledge distillation from the dense teacher, and parameter-efficient LoRA fine-tuning.

Table 7 reports zero shot task accuracy before and after recovery at 80% sparsity with the sparsity mask fixed throughout training. Fine tuning and distillation provide little improvement for the competing baselines. In several cases, performance decreases after recovery. In contrast, CWS responds more favorably to post pruning training. Distillation improves average accuracy from .301 to .314, exceeding the best zero shot competitor result of .309 from SparseGPT. The largest gain comes from combining distillation with LoRA, which increases average accuracy to .370. LoRA provides additional trainable parameters outside the frozen sparse weights, allowing recovery without

Table 5: HGRN-1.3B at 80% sparsity. PPL on WikiText-2. Downstream accuracies from zero-shot lm-eval.

Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	LMBD	Avg↑
Dense baseline	11.8	.510	.275	.480	.712	.528	.383	.481
Wanda	76,051	.260	.290	.257	.487	.519	.000	.302
RIA	28,615	.250	.302	.256	.513	.498	.000	.303
SparseGPT	6,956	.274	.289	.260	.511	.519	.000	.309
AWP	17,195	.260	.288	.265	.496	.507	.000	.303
CWS	131.27	.293	.232	.274	.520	.510	.004	.366

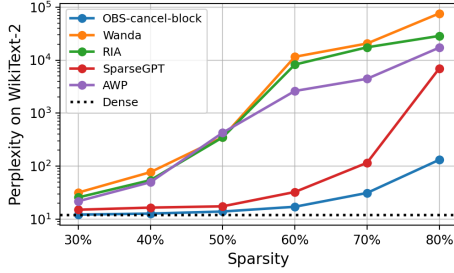


Figure 2: Sparsity-vs-perplexity. Sparsity sweep on HGRN-1.3B evaluated on WikiText-2.

Sparsity	CWS	Wanda	RIA	SGPT	AWP
30%	12.14	31.64	25.6	15.02	21.7
40%	12.69	76.87	54.3	16.46	49.7
50%	13.88	350	348	17.4	426.2
60%	17.07	11,552	8,239	32.43	2,616
70%	31.11	20,592	17,457	115.4	4,440
80%	131.27	76,051	28,615	6,956	17,195

Table 6: HGRN-1.3B sparsity sweep. PPL on the WikiText-2 test set. The lower PPL values are favorable.

Table 7: Post-pruning recovery for CWS at 80% sparsity on HGRN-1.3B. The sparsity mask is frozen throughout recovery. Downstream accuracies from zero-shot lm-eval.

Recovery	ARC-e	ARC-c	Hella.	PIQA	Wino.	LMBD	Avg↑
CWS	.263	.292	.250	.503	.489	.000	.301
CWS + FT	.286	.259	.265	.526	.521	.001	.310
CWS + Distil	.298	.259	.265	.526	.521	.001	.314
CWS + FT + LoRA	.290	.201	.263	.539	.486	.001	.356
CWS + Distil + LoRA	.309	.218	.265	.551	.508	.001	.370
Wanda	.260	.290	.257	.487	.519	.000	.302
Wanda + FT	.271	.294	.255	.502	.506	.000	.305
Wanda + Distil	.266	.259	.255	.502	.511	.000	.299
RIA	.250	.302	.256	.513	.498	.000	.303
RIA + FT	.273	.288	.258	.500	.506	.000	.304
RIA + Distil	.267	.278	.256	.499	.493	.000	.299
SparseGPT	.274	.289	.260	.511	.519	.000	.309
SparseGPT + FT	.279	.263	.257	.504	.480	.000	.297
SparseGPT + Distil	.280	.248	.264	.521	.502	.000	.302
AWP	.260	.288	.265	.496	.507	.000	.303
AWP + FT	.266	.271	.258	.497	.498	.000	.298
AWP + Distil	.269	.280	.260	.498	.490	.000	.300

modifying the sparsity pattern. These results suggest that CWS preserves more recoverable structure after one shot pruning than the competing methods.

3.5 LLaMA-7B

We further evaluate on the LLaMA-7B [Touvron et al., 2023], to analyze the effect of our pruning method in larger scale. Tables 8 and 9 report results for 50% and 80% pruned models, respectively. Table 10 and Figure 3 illustrate results from 30% to 80% sparsity. All accuracies are zero-shot.

Table 8: LLaMA-7B at 50% sparsity: task accuracy breakdown. PPL on WikiText-2 test set. Downstream accuracies from zero-shot lm-eval.

Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
Dense baseline	6.61	.678	.385	.694	.763	.675	.639
Wanda	8.69	.602	.347	.628	.732	.663	.595
RIA	8.58	.601	.332	.624	.732	.649	.588
SparseGPT	9.51	.609	.363	.649	.732	.643	.599
AWP	8.10	.614	.336	.612	.732	.661	.591
CWS	8.24	.610	.342	.631	.724	.658	.593

As depicted in Figure 3, at 7B parameters, the relative ordering of methods inverts with respect to the sub-2B results. AWP achieve lower perplexity, highlighting a tradeoff between solution quality

Table 9: LLaMA-7B at 80% sparsity: task accuracy breakdown. PPL on WikiText-2 test set. Downstream accuracies from zero-shot lm-eval.

Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
Dense baseline	6.61	.678	.385	.694	.763	.675	.639
Wanda	1,647.4	.253	.267	.265	.499	.491	.355
RIA	2,084.1	.268	.268	.261	.495	.496	.357
SparseGPT	2,071.2	.255	.265	.266	.513	.505	.361
AWP	211.1	.268	.263	.262	.503	.478	.355
CWS	245.5	.259	.270	.263	.497	.489	.356

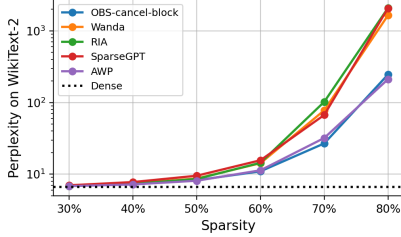


Figure 3: Sparsity-vs-perplexity. Sparsity sweep on LLaMA-7B evaluated on WikiText-2.

Sparsity	CWS	Wanda	RIA	SGPT	AWP
30%	6.87	6.89	6.84	6.99	6.81
40%	7.30	7.40	7.30	7.76	7.19
50%	8.24	8.69	8.58	9.51	8.10
60%	10.96	14.22	14.40	15.64	11.33
70%	26.85	77.82	102.17	67.02	31.99
80%	245.5	1,647.4	2,084.1	2,071.2	211.1

Table 10: Sparsity sweep on the LLaMA-7B. PPL on WikiText-2 test set.

and computational cost (Table 10). Despite AWP’s perplexity advantage, CWS retains the best zero-shot task accuracy at 80% sparsity (.362 vs. AWP .355). This reversal indicates that at larger scale, iterative gradient-based mask search navigates the high-sparsity regime more effectively than second-order one-shot methods. This improvement comes at a significantly higher computational cost. AWP requires up to 200 gradient-based updates within each layer, whereas CWS operates with a single pass over activation statistics and closed-form correction.

3.6 Ablation Studies

3.6.1 Selection Criterion

As shown in Table 11, mixing cancellation-aware scoring (r_j^2/d_j) with SparseGPT’s OBS corrections (Hybrid/Interleaved, PPL 587/786) performs worse than either method alone. SparseGPT’s correction is internally consistent with its own saliency score, and substituting on an external criterion breaks this coupling, propagating mismatch across layers. The fix is to derive cancellation-awareness from within the OBS objective itself, which is what our r_j^2/d_j criterion achieves. The global variant ($b \rightarrow \infty$, PPL 16.78) confirms the theoretical upper bound but causes Cholesky failures on LLaMA-7B and HGRN-1.3B, motivating the block-level design.

Table 11: Selection criterion ablation on the 1B transformer at 50% sparsity (WikiText-2 calibration). SparseGPT is included as the reference second-order baseline. Dense baseline PPL: 14.2.

Variant	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
SparseGPT (reference)	20.7	.504	.294	.428	.666	.517	.482
Greedy-cov (no correction)	635	.255	.300	.265	.499	.500	.364
Greedy+corr (batch correction)	316	.282	.260	.277	.502	.476	.360
Hybrid: cancellation + OBS correction	587	.288	.267	.283	.514	.470	.364
Interleaved: block cancel. + OBS	786	.287	.262	.266	.510	.492	.363
Global CWS ($b \rightarrow \infty$)	16.78	.490	.295	.430	.655	.504	.475
CWS (ours, $b = 128$)	18.06	.518	.288	.420	.663	.512	.480

171 4 CPU Inference

Sparse kernel. The model dimension d_{in} is partitioned into $d_g = d_{\text{in}}/G$ non-overlapping groups of size $G \in 32, 64, 128$. For each group g and output neuron j , the $K = G/4$ weight elements with largest absolute value are selected (25% density):

$$\mathcal{S}_{g,j} = \underset{i \in [gG, (g+1)G)}{\operatorname{argtopk}} |W_{i,j}|$$

172 The sparse Matrix-Vector Multiplication (GEMV) kernel computes:

$$y_j = \sum_{i \in \mathcal{S}_{g,j}}^{d_g} W_{i,j} \cdot x_i$$

173 Weights and their intra-group indices (uint8) are stored in a pre-interleaved format that enables
174 contiguous 512-bit loads aligned to group boundaries, keeping the working set for each group in L1
175 cache.

176 **Overflow weights fix-up.** The overflow set for group g and output j comprises the $G - K$ elements
177 not captured by the main kernel are collected across all groups and stored in a separate tensor:

$$\mathcal{O}_{g,j} = \{i \in [gG, (g+1)G) : i \notin \mathcal{S}_{g,j}, W_{i,j} \neq 0\}$$

178 The contribution of overflow weights is applied after the main kernel as a sparse matrix–vector
179 product:

$$y_j += \sum_{i \in \mathcal{O}_{g,j}} W_{i,j} \cdot x_i$$

180 **Hardware.** All benchmarks use Dual-socket AMD EPYC 9335 (Genoa, 4th-generation EPYC) and
181 16 OpenMP threads, keeping all threads on a single Non-uniform memory access (NUMA) domain
182 and avoiding inter-socket bandwidth contention. The CPU supports 512-bit extensions: AVX-512F,
183 AVX-512BW, AVX-512DQ, and AVX-512BF16.

Table 12: CPU inference benchmark for HGRN-1.3B at 80% sparsity.

Kernel	weight format	ms/token↓	speedup↑
Dense baseline (our)	BF16	27.1	1.00
PyTorch (reference)	FP32	266.0	0.1
Sparse ($G = 64$)	BF16	18.3	1.48
Sparse ($G = 128$)	BF16	20.6	1.32
Sparse ($G = 64$, no fix-up)	BF16	13.8	1.96
Sparse ($G = 128$, no fix-up)	BF16	12.1	2.23

184 5 Discussion

185 The experiments reveal a scale-dependent transition in what governs pruning quality. Two distinct
186 sources of perplexity degradation are the quality of the mask (did we remove the right weights?), and
187 the magnitude of the residual error (were the surviving weights corrected properly after removal?).
188 Their relative importance shifts with model scale and sparsity level. At the 1B to 1.3B scale,
189 residual error dominates once sparsity reaches moderate levels. Methods without exact second order
190 correction degrade rapidly at 50% sparsity, while CWS and SparseGPT remain stable. This suggests
191 that preserving the layer output after pruning is more important than improving the mask itself in the
192 sub 2B regime. The cancellation aware selection criterion provides an additional improvement by
193 identifying correlated channels whose outputs partially cancel under the activation distribution.

194 At 7B scale, parameter redundancy reduces per-layer residual error to recoverable levels at moderate
195 sparsity, shifting the bottleneck to mask quality. Once residual error is no longer catastrophic, more
196 exhaustive search over the combinatorial mask space pays off, which explains why AWP’s iterative
197 gradient-based search outperforms second-order one-shot methods on LLaMA-7B. At the same time,
198 the computational cost increases substantially. CWS requires a single pass over activation statistics
199 and closed form correction, while AWP performs repeated gradient updates within each layer.

200 The CPU benchmarks show that the sparsity patterns produced by CWS remain useful during
201 inference. Using AVX-512 sparse kernels, the pruned HGRN-1.3B models achieve close to $2\times$ end
202 to end speedup over dense BF16 inference. Further more, recovery experiments at 80% sparsity show
203 that the pruned models still retain useful structure after one shot pruning. Distillation with LoRA
204 improves average accuracy from 0.301 to 0.370 without modifying the sparsity mask. Competing
205 methods show much smaller gains after recovery.

206 CWS currently applies greedy selection within fixed width blocks to maintain numerical stability
207 during the Schur complement updates. Larger or adaptive blocks may capture additional cancellation
208 structure, particularly on wider models. The current evaluation covers transformer and recurrent SSM
209 architectures at 1B to 7B scale. Extending the method to mixture of experts models and larger scales
210 remains future work.

References

- Elias Frantar and Dan Alistarh. Sparsegpt: Massive language models can be accurately pruned in one-shot. In *International conference on machine learning*, pages 10323–10337. PMLR, 2023.
- Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac’h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. The language model evaluation harness, 2024. URL <https://zenodo.org/records/12608602>.
- Song Han, Jeff Pool, John Tran, and William Dally. Learning both weights and connections for efficient neural network. *Advances in neural information processing systems*, 28, 2015.
- Babak Hassibi and David Stork. Second order derivatives for network pruning: Optimal brain surgeon. *Advances in neural information processing systems*, 5, 1992.
- Jing Liu, Toshiaki Koike-Akino, Ye Wang, Hassan Mansour, and Matthew Brand. Awp: Activation-aware weight pruning and quantization with projected gradient descent. *arXiv preprint arXiv:2506.10205*, 2025.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. *arXiv preprint arXiv:1609.07843*, 2016.
- Zhen Qin, Songlin Yang, and Yiran Zhong. Hierarchically gated recurrent neural network for sequence modeling. *Advances in Neural Information Processing Systems*, 36:33202–33221, 2023.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020.
- Mingjie Sun, Zhuang Liu, Anna Bair, and J Zico Kolter. A simple and effective pruning approach for large language models. *arXiv preprint arXiv:2306.11695*, 2023.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Yingtao Zhang, Haoli Bai, Haokun Lin, Jialin Zhao, Lu Hou, and Carlo Vittorio Cannistraci. Plug-and-play: An efficient post-training pruning method for large language models. In *The Twelfth International Conference on Learning Representations*, 2024.

A Additional Ablation Studies

A.0.1 Calibration Data

As shown in Table 13, with WikiText-2 calibration, SparseGPT leads on task accuracy (0.492 vs. 0.465). Switching to C4 reverses this (CWS: 0.443 vs. 0.435, PPL 18.6 vs. 21.6). We attribute SparseGPT’s sensitivity to its fixed left-to-right column ordering. OBS corrections are most effective when the calibration distribution matches the pruning order. CWS selects on cancellation structure (a property of geometry), giving more consistent behavior across calibration sources. AWP, Wanda, and RIA are largely insensitive to calibration choices, as expected when reconstruction error dominates or no weight correction is applied.

Table 13: Effect of calibration data on the 1B transformer at 50% sparsity. PPL on WikiText-2. Downstream accuracies from zero-shot lm-eval.

Calib. data	Method	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
WikiText-2	Wanda	3,048	.271	.285	.263	.502	.493	.363
	RIA	1,729	.277	.263	.268	.513	.490	.362
	SparseGPT	20.7	.504	.294	.428	.666	.517	.482
	AWP	241.8	.332	.244	.285	.528	.510	.380
	CWS	18.06	.518	.288	.420	.663	.512	.480
C4	Wanda	3,492	.271	.275	.267	.502	.489	.360
	RIA	1,897	.270	.267	.272	.515	.496	.364
	SparseGPT	21.6	.505	.298	.429	.678	.517	.485
	AWP	290.1	.334	.251	.283	.530	.507	.381
	CWS	18.2	.519	.291	.426	.663	.541	.488

A.1 Block Size Sweep

Table 14 shows that increasing the OBS block size consistently improves perplexity, as larger blocks capture longer-range cancellation interactions across columns. The largest improvement occurs from $b = 64$ to $b = 128$ (18.60 $\rightarrow$ 18.06), while block-sizes larger than 128 provide only marginal gains (128 $\rightarrow$ 256: -0.03 , 256 $\rightarrow$ 512: -0.26).

However, downstream zero-shot accuracy does not improve monotonically with larger blocks. While moderate block sizes (128–256) achieve the best average accuracy, larger blocks slightly degrade downstream performance despite lower perplexity. This suggests that increasingly global OBS corrections may overfit the calibration objective without consistently preserving task-level behavior.

Importantly, block size 128 is the largest configuration that remains numerically stable across all evaluated model families. Larger blocks (256 and 512) occasionally produce Cholesky failures on wider layers of LLaMA-7B and HGRN-1.3B due to ill-conditioned covariance sub-matrices. Therefore, we use $b = 128$ throughout the paper as the best trade-off between perplexity, downstream accuracy, computational cost, and numerical stability.

Table 14: Block size sweep on the 1B transformer at 50% sparsity. PPL on WikiText-2. Downstream accuracies from zero-shot lm-eval. Dense baseline PPL: 14.2.

Block size b	PPL↓	ARC-e	ARC-c	Hella.	PIQA	Wino.	Avg↑
Dense	14.20	.530	.306	.451	.692	.525	.501
64	18.60	.519	.283	.420	.652	.519	.479
128 (ours)	18.06	.518	.289	.421	.661	.517	.481
256	18.03	.510	.290	.421	.671	.526	.480
512	17.77	.503	.278	.429	.667	.513	.478
∞ (global CWS)	16.78	.490	.295	.430	.655	.504	.475

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